

The Agony and the Ecstasy

Constructing a "Crash-Filtered" Equity Index using Machine Learning

Tristan Leiter

Vienna University of Business and Economics

Supervisor: Prof. Kurt Hornik

May 25, 2026

The Agony and Ecstasy

Passive indices are exposed to both extremes

Long-run index returns are driven by a few high-performing stocks, while a considerable share of constituents suffer catastrophic declines.

Ecstasy

- Extreme winners carry a large share of long-run wealth creation.
- Winners can be volatile, expensive, or distressed.
- Thus, even a small amount of disregarded winners can hurt performance.

Agony

- Some constituents experience deep and persistent drawdowns without recovery.
- Passive strategies may hold imploding firms for too long.

Solution

The challenge is not to eliminate highly volatile stocks. It is to distinguish recoverable volatility from persistent implosion. The empirical question is whether catastrophically declining firms are forecastable using ideas from credit-risk modelling, but with market-based distress signals.

Research Question

Main research question

To what extent does a “Crash-Filtered” equity index, constructed via probabilistic implosion modeling, generate superior risk-adjusted returns compared to the market benchmark and traditional minimum-volatility strategies?

The probabilistic implosion model follows the Catastrophic Stock Implosion approach proposed by Tewari et al. (2024). Instead of relying only on formal bankruptcy declarations, CSI uses market-based price-path signals.

Subquestions:

- How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?
- To what extent do Ensemble methods reduce the False Positive Rate (classifying recoverable volatility as CSI) compared to traditional volatility-based exclusion strategies, while maintaining Recall?
- Which features are most important for distinguishing between “Zombie” firms (CSI) and non-zombie firms?

Empirical design

The main data sources are WRDS and FRED. The OOS index testing is kept strictly separated from the training data for walk-forward CV and the test set. Model selection is conducted based on the test set results.

Data sources

CRSP

monthly returns, market cap, delistings, wealth paths

Compustat

annual accounting fundamentals

FRED

macro variables

Universe

Minimum size

market cap $\geq$ USD 100M

Size buckets

CRSP-like Total Market, Large, Mid, Small Cap

~

Sample period

Coverage

Jan 1993 – Dec 2024 (32y, monthly frequency)

Cross-validation

3 expanding folds

seed = 123

Temporal split



Methodology I: Response Variable Classification

Following Tewari et al. (2024), a stock is classified as a CSI ($y = 1$) if it satisfies all three criteria:

- **Initial Crash (C):** more than a C drawdown from the trailing peak marks the beginning of the “zombie” period. $C \leq -80\%$
- **Non-Recovery (M):** maximum cumulative return remains below the recovery ceiling M during the zombie period. $M \leq -20\%$
- **Zombie Period (T):** duration of the confirmation window during which non-recovery must hold. $T = 18 \text{ months}$

Tewari et al. (2024) use a yearly prediction horizon ($h = 12$ months):

$$y_{i,t} = \begin{cases} 1, & \text{if stock } i \text{ triggers a crash } C \text{ within } [t, t+h] \text{ and the zombie criteria are met,} \\ 0, & \text{otherwise.} \end{cases}$$

Implementation

Events are detected monthly from CRSP wealth paths and then aligned to annual feature rows: a trigger in calendar year $t + 1$ gives the firm-year observation at t a positive label.

Applying the classification to the CRSP dataset

Classification

The current cleaned scaffold keeps only observable CRSP firm-years. With the terminal-failure route enabled, the temporary-CSI label has 8,517 positive annual cells and 8,674 unresolved cells retained as $y = NA$.

Split	Obs. rows	$y = 0$	$y = 1$	$y = NA$	Labelled	Prev.
Full	188,460	171,269	8,517	8,674	179,786	4.74%
Train	143,173	136,630	6,543	0	143,173	4.57%
Test	18,111	17,307	804	0	18,111	4.44%
OOS	27,176	17,332	1,170	8,674	18,502	6.32%

Motivation for robustness

Prevalence is computed on labelled cells only: $y = 1/(y = 0 + y = 1)$. The retained $y = NA$ rows are observable firm-years with unresolved labels, not deleted observations.

Robustness I: Do CSI-firms go bankrupt?

Validation question: Among firms that eventually delist or bankrupt, how many had already been classified as CSI?

Scenario	CSI rows	CSI firms	CRSP 572-574 firms	Detected	Missed	Detection rate
Before: confirmed_csi only	8,369	5,642	629	151	478	24.0%

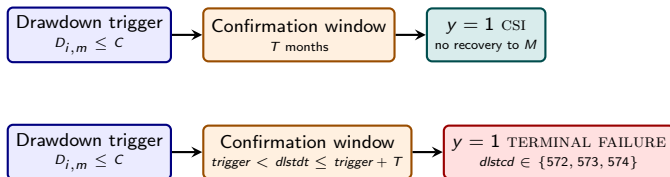
CRSP bankruptcy-related delisting codes 572, 573, and 574 are used as the terminal-distress audit set.

Core issue: The low-detection rate causes methodological issues. If one sticks strictly to the paper, then the CSI methodology should catch firms which do not recover anymore. Yet, the data shows a different result.

Methodology II: “Temporary-CSI”

Revision

Original CSI rule unchanged. A second positive route is added only after a valid drawdown trigger: if CRSP 572–574 arrives after the trigger and within the T -month confirmation window, the trigger becomes positive.



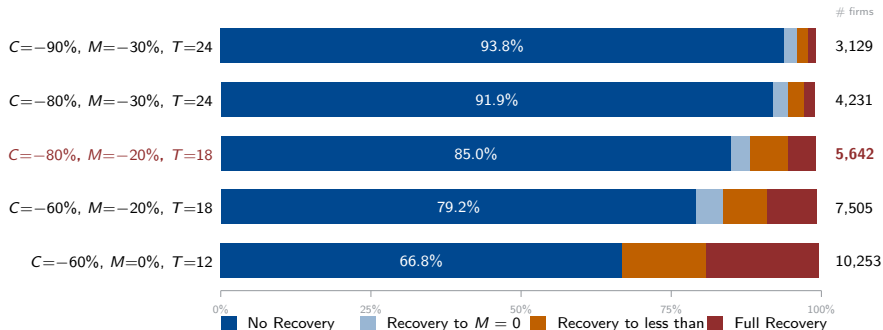
Scenario	Bankrupting	Detected	Missed	Detection
Before: CSI only	629	151	478	24.0%
After: CSI + terminal failure	629	545	84	86.65%

The 84 remaining misses are treated as methodological exclusions under the retained CSI-trigger rule, not as known pipeline errors.

Robustness II: Do CSI-firms recover again?

CSI flags mostly identify persistent declines; recovery risk appears mainly in with a smaller drawdown threshold

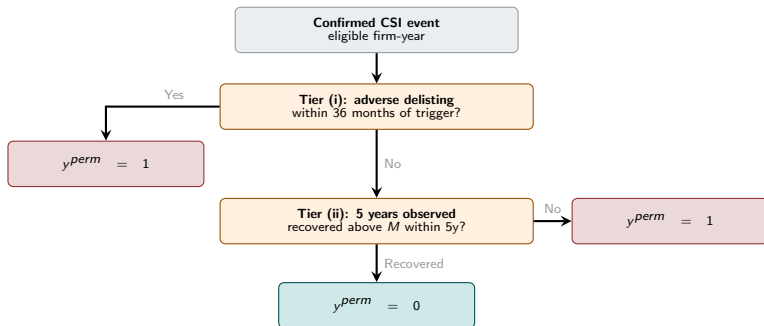
After a temporary-CSI flag, the robustness grid tracks the five-year recovery bucket across alternative crash/recovery windows.



Methodology III: “Permanent-CSI”

Dealing with recovering firms

Permanent-CSI starts from a confirmed temporary-CSI event and separates terminal failure, durable non-recovery, resolved recovery, and unresolved PCL windows.



Contrast with temporary CSI

Temporary CSI applies a finite lockout after a flag. Permanent-CSI marks terminal failure or durable non-recovery as $y^{perm} = 1$, resolved recovery as $y^{perm} = 0$, and unresolved PCL windows as $y = \text{NA}$. Current prevalence: 6,258 positives across 187,626 labelled firm-years (3.34%).

Dataset: Descriptive Statistics

CSI positives are rare and economically distinct from non-positives

Both label definitions isolate a small distressed tail. The key contrast is not only prevalence: positive firms are much smaller and show worse realized returns and profitability.

Sample	CSI	y	Obs.	Share	Firms	Median mcap	Ann. ret.	ROA
Full	Temp.	0	171,269	95.26%	19,073	254.9	4.9%	1.6%
Full	Temp.	1	8,517	4.74%	5,603	74.9	-35.0%	-18.8%
Full	Perm.	0	181,368	96.66%	19,565	252.6	4.0%	1.5%
Full	Perm.	1	6,258	3.34%	4,696	65.7	-37.5%	-20.0%

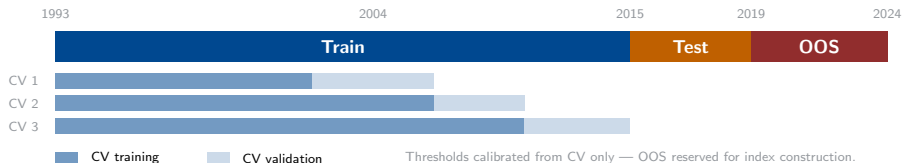
Read

The full temporary-CSI classification contains **8,517** positive annual cells. Positive labels identify much smaller firms with sharply negative returns and profitability.

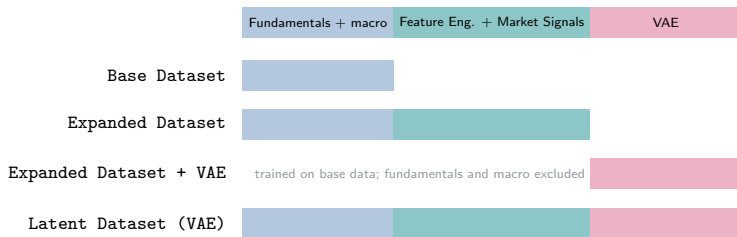
Remark: Obs. are annual label cells; Firms are unique firms. Market cap is USD millions.

Modelling I: Setup and Feature Engineering

Validation design



Feature engineering and feature-set keys



Modelling II: Results Temporary CSI

Best current model for index construction

AutoGluon Expanded Dataset is the best current temporary-CSI model on CV and the 2016-2019 test set OOS portfolio testing is deliberately kept out of model selection.

Model	CV AP	CV AUC	CV R3	Test AP	Test AUC	Test R3
AG Expanded Dataset	0.2197	0.9647	0.6127	0.2114	0.9692	0.6551
AG Latent Dataset (VAE)	0.1639	0.9531	0.5110	0.1433	0.9549	0.5130
XGB Expanded Dataset	0.2094	0.9525	0.5293	0.1813	0.9670	0.6391
XGB Exp. Dataset + VAE	0.2080	0.9463	0.5275	0.1837	0.9671	0.6279
XGB Latent Dataset (VAE)	0.1658	0.9406	0.4499	0.1462	0.9553	0.5179

Interpretation

The initial question is answered positively: temporary CSI is forecastable, but the expanded dataset derived from feature engineering and market signals is essential because Expanded Dataset + VAE alone is materially weaker. For index construction, AG Expanded Dataset is the clean model-selection baseline, with XGBoost as a transparent robustness challenger. False positives remain the economic constraint, so thresholds should be calibrated from CV-FPR targets before any OOS portfolio evaluation.

Remark: R3 = recall at $\text{FPR} \leq 3\%$. Full AutoGluon and XGBoost temporary-CSI tables are in Appendix A10-A11.

Modelling III: Results Permanent-CSI

Best current model evidence

Permanent-CSI is forecastable but more difficult than temporary CSI. **XGBoost** Expanded Dataset is strongest on 2016-2019 test AP/R3; **AutoGluon** Expanded Dataset and Exp. Dataset + VAE are the clean completed AutoGluon baselines.

Model	CV AP	CV AUC	CV R3	Test AP	Test AUC	Test R3
AG Expanded Dataset	0.2016	0.9650	0.6216	0.1480	0.9674	0.6621
AG Exp. Dataset + VAE	0.2019	0.9651	0.6169	0.1471	0.9670	0.6655
XGB Expanded Dataset	0.1928	0.9462	0.5498	0.1549	0.9681	0.6707
XGB Exp. Dataset + VAE	0.1947	0.9432	0.5501	0.1393	0.9682	0.6398
AG Latent Dataset (VAE)	0.1529	0.9573	0.5455	0.1190	0.9600	0.5592

Interpretation

The permanent target has usable ranking structure, but AP is materially lower than for temporary CSI because positives are rarer. Engineered Expanded Dataset features remain the core signal; latent-only evidence is weaker on CV/test even though it improves on test-OOS AP. For index construction, permanent CSI should be treated as a stricter removal signal and benchmarked with conservative FPR thresholds.

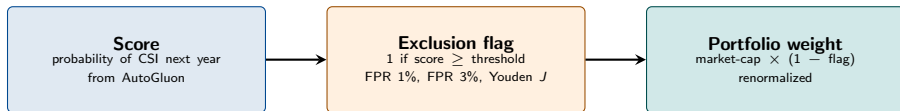
Remark: R3 = recall at $\text{FPR} \leq 3\%$. Full permanent-CSI XGBoost and AutoGluon tables are in Appendix A12-A13.

Index Construction I: From Score to Portfolio Weight

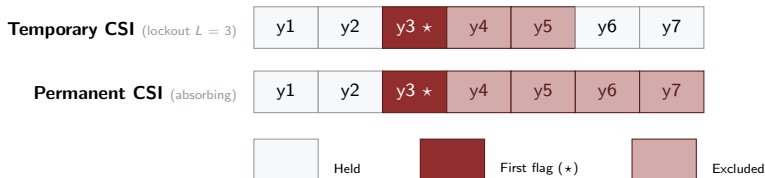
One score rule, two exclusion regimes

The model produces a probability of CSI next year. A threshold turns that probability into a binary exclusion flag. The flag is then applied to market-cap weights. Two regimes differ only in how long the flag persists after it first fires.

From model score to portfolio weight



How the flag evolves over time — a firm crosses the threshold in year 3



Index Construction II: Four Benchmark Universes

Same exclusion rule, four universes

The revised 11C track applies the score-to-weight pipeline independently inside each CRSP-like universe, producing four screened benchmarks alongside their unfiltered counterparts.

Total Market

full universe
≥ USD 100M

Large Cap

CRSP-like
top tier

Mid Cap

CRSP-like
middle tier

Small Cap

CRSP-like
bottom tier

Benchmark construction

- Universe formed from prior-December market capitalization.
- Minimum market cap: USD 100 million.
- Quarterly rebalancing: March, June, September, December.
- Market-cap weights; prototype transaction costs are 0 bps.

Signal discipline

- AutoGluon Expanded Dataset score is annual.
- Holding-year signal uses information available before the holding year.
- Thresholds calibrated from CV predictions only
- never from OOS returns.

Index Construction Results: Temporary CSI Performance

Test-OOS best rules by universe: benchmark row followed by filtered strategy

Universe	Strategy	Geo ret.	Ann. SD	Sharpe	Max DD	ES 2.5%	Δ pp
Total Market	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	-
	FPR3 / 5yr	13.41%	18.50%	0.615	-24.20%	-11.44%	+0.119
Large Cap	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	-
	FPR1 / 1yr	14.26%	18.49%	0.656	-24.79%	-10.83%	-0.020
Mid Cap	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	-
	Youden / 1yr	9.68%	20.09%	0.414	-23.55%	-13.03%	+0.030
Small Cap	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	-
	Youden / 3yr	9.03%	21.98%	0.370	-29.12%	-15.60%	+1.372

Remark: Δ is the annualized geometric-return difference vs benchmark.

Index Construction Results: Temporary CSI – Error Costs

Where the winning temporary strategies benefit

The filtered portfolios give up return through false positives, but offset this mainly through retained true-negative exposure. Small Cap has the largest return improvement and also the largest false-positive drag.

Universe	Best rule	FP cost	FN cost	TP gain	TN gain	Net
Total Market	FPR3 / 5yr	-0.970	-0.003	+0.064	+1.028	+0.119
Large Cap	FPR1 / 1yr	-0.018	+0.000	+0.000	-0.002	-0.020
Mid Cap	Youden / 1yr	-3.552	+0.000	-0.004	+3.586	+0.030
Small Cap	Youden / 3yr	-7.648	-0.034	+0.463	+8.591	+1.372

Interpretation: The true-positive protection is positive but small. The key tradeoff is whether retained non-CSI exposure compensates for false-positive opportunity costs. *Remark:* Annualized pp terms. TN gain is the geometric-return balancing retained-exposure term, so $FP+FN+TP+TN$ equals Net.

Index Construction Results: Permanent CSI Performance

Test-OOS best rules by universe: benchmark row followed by filtered strategy

Universe	Strategy	Geo ret.	Ann. SD	Sharpe	Max DD	ES 2.5%	Δ pp
Total Market	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	—
	FPR3 / perm.	13.76%	18.64%	0.629	-24.21%	-11.38%	+0.472
Large Cap	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	—
	FPR3 / perm.	14.62%	18.30%	0.678	-24.10%	-10.73%	+0.345
Mid Cap	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	—
	FPR1 / perm.	10.08%	20.56%	0.427	-24.38%	-13.85%	+0.426
Small Cap	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	—
	FPR3 / perm.	7.61%	23.13%	0.306	-29.39%	-16.11%	-0.042

Remark: Δ is annualized geometric-return difference vs benchmark.

Index Construction Results: Permanent CSI - Error Costs

Where the winning permanent strategies benefit

Permanent CSI improves Total, Large, and Mid Cap primarily through true-negative retained exposure after absorbing removals. Small Cap remains weak because false-positive costs absorb the benefit.

Universe	Best rule	FP cost	FN cost	TP gain	TN gain	Net
Total Market	FPR3 / perm.	-1.261	-0.001	+0.022	+1.712	+0.472
Large Cap	FPR3 / perm.	-0.849	+0.000	+0.000	+1.194	+0.345
Mid Cap	FPR1 / perm.	-0.167	+0.000	+0.000	+0.593	+0.426
Small Cap	FPR3 / perm.	-3.109	-0.017	+0.076	+3.008	-0.042

Read: the strongest permanent-CSI case is not direct true-positive protection; it is disciplined removal that preserves exposure to retained non-CSI firms while avoiding enough weak names. *Remark:* Annualized pp terms. TN gain is the geometric-return balancing retained-exposure term, so $FP+FN+TP+TN$ equals Net.

Next steps before the final presentation

Four key areas

The proposal lays the methodological and empirical foundation. Remaining work splits into data validation, model interpretability, an optional modeling extension, and an index-construction extension - each connected to a section already in the proposal.

Data validation

Document residual misses

Current audit leaves 84 CRSP 572–574 firms outside revised temporary CSI; a separate diagnostic can decompose these methodological exclusions if needed.

Extends Methodology II audit.

Model interpretability

Feature importance analysis

Feature importance attribution as in the OeNB credit-risk modelling presentation, applied to AutoGluon Expanded Dataset and Exp. Dataset + VAE.

Answers RQ 3 directly.

Modeling extension

LightGBM with Bayesian HPO

Transparent challenger to AutoGluon with hyperparameter optimization via Bayesian optimization.

Optional, time permitting.

Index construction extension

Low-vol & quality variants

Re-use the CSI score to build a Low-Vol benchmark; test whether retained-firm selection works as a quality substitute (lowest-risk firms instead of excluding the highest).

Extends Index Construction track.

Appendix A1: Recovery Buckets Across C/M/T Grid

Base grid C080_M020_T018: stayed below $M = -20\%$: 4,793 firms (85.0%); recovered M to 0: 182 (3.2%); recovered 0 to $|C|$: 354 (6.3%); recovered $> |C|$: 260 (4.6%); no follow-up: 53 (0.9%).

Grid	Firms	Stayed below M	M to 0	0 to $ C $	$> C $
C090_M030_T012	3,596	87.2%	3.8%	4.2%	3.8%
C090_M030_T018	3,316	91.6%	2.9%	2.7%	1.8%
C090_M030_T024	3,129	93.8%	2.2%	1.9%	1.2%
C090_M020_T012	4,437	82.3%	3.4%	6.7%	6.6%
C090_M020_T018	4,125	88.4%	2.7%	4.5%	3.6%
C090_M020_T024	3,911	91.5%	2.1%	3.1%	2.4%
C090_M000_T012	5,644	81.3%	0.0%	8.6%	9.4%
C090_M000_T018	5,350	86.8%	0.0%	6.2%	6.1%
C090_M000_T024	5,061	90.5%	0.0%	4.6%	4.0%
C080_M030_T012	4,843	83.2%	5.1%	5.6%	5.1%
C080_M030_T018	4,483	88.7%	3.7%	4.0%	2.5%
C080_M030_T024	4,231	91.9%	2.6%	2.7%	1.7%
C080_M020_T012	6,073	78.1%	3.8%	8.8%	8.5%
C080_M020_T018	5,642	85.0%	3.2%	6.3%	4.6%
C080_M020_T024	5,348	88.9%	2.5%	4.5%	3.1%
C080_M000_T012	7,562	74.5%	0.0%	12.1%	12.8%
C080_M000_T018	7,096	81.5%	0.0%	9.5%	8.2%
C080_M000_T024	6,760	85.8%	0.0%	7.2%	6.0%
C060_M030_T012	6,304	78.3%	6.8%	6.3%	7.8%
C060_M030_T018	5,850	84.3%	5.1%	4.7%	4.9%
C060_M030_T024	5,517	88.7%	3.7%	3.4%	3.1%
C060_M020_T012	8,068	71.9%	5.7%	9.6%	12.2%
C060_M020_T018	7,505	79.2%	4.4%	7.4%	8.3%
C060_M020_T024	7,098	84.5%	3.6%	5.5%	5.6%
C060_M000_T012	10,253	66.8%	0.0%	14.1%	18.7%
C060_M000_T018	9,621	74.0%	0.0%	11.8%	13.7%
C060_M000_T024	9,170	79.7%	0.0%	9.3%	10.3%

Notes: Firms = unique flagged firms. Stayed below M = never recovered above recovery ceiling M . M to 0 = recovered above M but not to break-even. 0 to $|C|$ = recovered above break-even but not more than the absolute crash threshold. $> |C|$ = recovered more than the absolute crash threshold. Percentages may not sum to 100% because no-follow-up firms are omitted from the displayed bucket columns.

Appendix A2: Delisting and Bankruptcy Detection

Indicator group	CRSP firms	Detected	Rate	Positive rows followed
Bankruptcy range 572-574	629	545	86.65%	2,757
Bankruptcy / declared insolvent 574	548	513	93.61%	2,502
Adverse 400-490, 550-585	5,654	3,676	65.02%	7,000

Why the override is additive

The CRSP signal is not an independent default label. It only turns a CSI-like trigger into a positive `dynamic_csi` event when the firm fails before the 18-month confirmation clock can fully resolve.

Appendix A3: Revised Annual Prevalence

Current CRSP 572–574 overlap

Annual label metric		Metric	Count
	Count		
Temporary positive label cells	8,517	CRSP 572–574 firms in sample	629
Temporary unresolved cells ($y = NA$)	8,674	Detected by revised temporary	545
Permanent positive label cells	6,258	CSI	
Permanent unresolved cells ($y = NA$)	834	Missed by revised temporary CSI	84
		Detection rate	86.65%

The remaining misses are not treated as current pipeline errors. Under the additive rule, CRSP bankruptcy codes only validate an already triggered CSI event within the timing window; they are not standalone default labels.

Methodological status

The current overlap audit supports the 545/84/86.65% story. A full reason breakdown for all 84 CRSP 572–574 misses would require a separate diagnostic; the current slide treats them as exclusions under the retained trigger-and-window definition.

Appendix A4: Cleaned Label Counts

CSI type	Split	Obs. rows	$y = 0$	$y = 1$	$y = NA$	Labelled	Prev.
Temporary	Full	188,460	171,269	8,517	8,674	179,786	4.74%
Temporary	Train	143,173	136,630	6,543	0	143,173	4.57%
Temporary	Test	18,111	17,307	804	0	18,111	4.44%
Temporary	OOS	27,176	17,332	1,170	8,674	18,502	6.32%
Permanent	Full	188,460	181,368	6,258	834	187,626	3.34%
Permanent	Train	143,173	137,794	5,379	0	143,173	3.76%
Permanent	Test	18,111	17,511	542	58	18,053	3.00%
Permanent	OOS	27,176	26,063	337	776	26,400	1.28%

Read

Both targets remain rare after the observable-scaffold cleanup. $y = NA$ rows stay in canonical artifacts but are excluded from supervised training and label-based evaluation.

Source: Labelled rows = $y = 0 + y = 1$; prevalence = $y = 1$ /labelled rows.

Appendix A5: Descriptive Statistics – Market Cap

Market cap, USD millions

CSI type	Split	y	Mean	Median	P25	P75
Temporary	CV	0	4,117.4	408.1	87.3	1,924.5
Temporary	CV	1	613.4	85.1	30.0	255.5
Temporary	Full	0	4,864.6	350.9	67.2	1,903.7
Temporary	Full	1	574.6	68.7	23.3	228.0
Permanent	CV	0	3,452.1	354.5	85.2	1,551.6
Permanent	CV	1	447.3	79.6	30.3	221.6
Permanent	Full	0	3,558.3	249.3	57.7	1,232.6
Permanent	Full	1	496.5	66.2	22.7	219.0

Read

CSI-positive firms are much smaller across both definitions. In the full sample, median market cap is 68.7 for temporary positives versus 350.9 for non-positives, and 66.2 for permanent positives versus 249.3 for non-positives.

Source: Market cap is USD millions.

Appendix A6: Descriptive Statistics – Other Medians

CSI type	Split	y	Ann. ret.	Assets	Sales	ROA	M/B
Temporary	CV	0	6.1%	640.9	326.7	1.7%	1.82
Temporary	CV	1	-24.9%	86.1	50.2	-14.2%	1.51
Temporary	Full	0	4.9%	565.3	295.2	1.6%	1.81
Temporary	Full	1	-35.0%	77.5	46.8	-18.8%	1.59
Permanent	CV	0	6.0%	567.8	270.2	1.7%	1.82
Permanent	CV	1	-28.3%	78.8	44.4	-15.8%	1.46
Permanent	Full	0	4.2%	395.3	191.9	1.5%	1.80
Permanent	Full	1	-37.5%	64.6	35.1	-19.3%	1.52

Read

The positive class has sharply lower realized returns, smaller operating scale, and negative profitability. The contrast strengthens in the full sample for both temporary and permanent CSI.

Source: Assets and sales are USD millions.

Appendix A7: CV Folds, Leakage Controls, and Training Metric

Expanding-window CV

Fold	Training years	Validation years
1	Initial training block	No validation; used to seed expanding window
2	Up to 1998	1999-2004
3	Up to 2004	2005-2009
4	Up to 2009	2010-2015

Training / selection metric

- **AutoGluon:** yes, model ranking uses AP via `EVAL_METRIC = "average_precision"` passed to `TabularPredictor(..., eval_metric=EVAL_METRIC)` in `09C_AutoGluon.py`.
- **XGBoost:** binary-logistic training uses `eval_metric="aucpr"` in `09_Train.R`; reported AP is computed from predicted probabilities.
- Split before preprocessing.
- Train-fitted winsorization, imputation, and PIT only.
- CV AP/AUC/R@FPR metrics are averaged per fold; OOS is reserved for index construction, not model selection.

Appendix A8: Feature Family Inventory

Family	Examples / purpose
Fundamentals	Balance sheet, income statement, profitability, leverage, liquidity, accounting dynamics.
Market and price	Returns, drawdowns, volatility, momentum, market capitalization, price-path information.
Macro	FRED macro series and macro interaction terms.
Latent	VAE latent factors from fundamentals or raw features plus reconstruction error.
Metadata excluded	Labels, event dates, confirmation dates, censoring flags, and diagnostics are retained for audit but excluded from model features.

Appendix A9: VAE Architecture Details

Component	Setting
Encoder	$n_features \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow z_mean, z_log_var$
Latent dimension	24
Decoder	$z \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow n_features$
Loss	Reconstruction + β KL + γ supervised CSI classification loss
Parameters	$\beta = 1.0$, $\gamma = 0.1$, early stopping patience 15
Status	Weakly supervised on training rows; test and OOS labels are not used for VAE fitting.

Appendix A10: AutoGluon Results for Temporary CSI

CV average Per-fold average across expanding-window folds 2–4.

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.2076	0.9589	0.5793	0.7588
raw	0.2197	0.9647	0.6127	0.7803
raw + latent	0.2173	0.9643	0.6002	0.7786
latent raw	0.1639	0.9531	0.5110	0.6900

Test: 2016–2019

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.1758	0.9550	0.6069	0.7726
raw	0.2114	0.9692	0.6551	0.8183
raw + latent	0.1874	0.9680	0.6354	0.8171
latent raw	0.1433	0.9549	0.5130	0.7095

OOS: 2020–2024

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.2565	0.9300	0.6220	0.7935
raw	0.2911	0.9552	0.7169	0.8401
raw + latent	0.2951	0.9577	0.7169	0.8385
latent raw	0.2720	0.9509	0.6436	0.8043

Main read

raw is best on the 2016–2019 test set by AP, AUC, and R@FPR3. raw + latent is very close on AUC/R@FPR thresholds and is best on OOS AP/AUC.

Appendix A11: XGBoost Results for Temporary CSI

Finalized temporary-CSI XGBoost rerun Full CV metrics stored.

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.1978	0.9330	0.5173	0.6847
raw	0.2094	0.9525	0.5293	0.6995
raw + latent	0.2080	0.9463	0.5275	0.6930
latent raw	0.1658	0.9406	0.4499	0.6167

Test: 2016–2019

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.1612	0.9568	0.5748	0.7812
raw	0.1813	0.9670	0.6391	0.8232
raw + latent	0.1837	0.9671	0.6279	0.8319
latent raw	0.1462	0.9553	0.5179	0.7330

OOS: 2020–2024

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.2643	0.9339	0.6520	0.7868
raw	0.2578	0.9473	0.6769	0.8260
raw + latent	0.2880	0.9494	0.6994	0.8393
latent raw	0.2723	0.9532	0.6644	0.8060

Main read

XGBoost confirms the same modelling pattern: raw + latent is strongest OOS by AP and R@FPR thresholds, while latent raw alone is not a substitute for engineered raw features.

Appendix A12: XGBoost Results for Permanent CSI

Permanent-CSI XGBoost results Same feature-set variants as temporary CSI.

Model	AP	AUC	R@FPR3	R@FPR5
raw	0.1928	0.9462	0.5498	0.7118
fund	0.1841	0.9268	0.5350	0.6942
raw_plus_latent	0.1947	0.9432	0.5501	0.7118
latent_raw	0.1578	0.9428	0.4852	0.6525

Test: 2016–2019

Model	AP	AUC	R@FPR3	R@FPR5
raw	0.1549	0.9681	0.6707	0.8336
fund	0.1262	0.9555	0.6003	0.7856
raw_plus_latent	0.1393	0.9682	0.6398	0.8388
latent_raw	0.1252	0.9601	0.5918	0.7839

Test-OOS: 2020–2024

Model	AP	AUC	R@FPR3	R@FPR5
raw	0.0514	0.9424	0.3767	0.6340
fund	0.0507	0.9249	0.4032	0.6446
raw_plus_latent	0.0550	0.9433	0.4085	0.6578
latent_raw	0.0718	0.9401	0.4748	0.6525

Main read

raw is strongest on the 2016–2019 test set by AP and R@FPR3, while raw_plus_latent is marginally strongest on test AUC/R@FPR5 and CV AP. Test-OOS is weaker and unusual: latent_raw has the highest AP/R@FPR3, so permanent-CSI XGBoost evidence is less stable than the temporary-CSI XGBoost pattern.

Appendix A13: AutoGluon Results for Permanent CSI

Permanent-CSI AutoGluon results Completed runs: fund, raw, raw_plus_latent.

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.1884	0.9584	0.5981	0.7702
raw	0.2016	0.9650	0.6216	0.7835
raw_plus_latent	0.2019	0.9651	0.6169	0.7890
latent_raw	0.1529	0.9573	0.5455	0.7212

Test: 2016–2019

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.1300	0.9527	0.6364	0.7856
raw	0.1480	0.9674	0.6621	0.8233
raw_plus_latent	0.1471	0.9670	0.6655	0.8199
latent_raw	0.1190	0.9600	0.5592	0.7496

Test-OOS: 2020–2024

Model	AP	AUC	R@FPR3	R@FPR5
fund	0.0521	0.9224	0.4324	0.6366
raw	0.0587	0.9495	0.4456	0.7109
raw_plus_latent	0.0581	0.9502	0.4668	0.6950
latent_raw	0.0696	0.9438	0.4695	0.6950

Main read

raw is the best AutoGluon permanent-CSI model on the 2016–2019 test set by AP and R@FPR5. raw_plus_latent is essentially tied on CV and has the best test-OOS AUC/R@FPR3, while latent_raw performs best on test-OOS AP but remains weaker on CV/test AP.

Appendix A14: Index Strategy Construction

Strategy menu

- **Benchmark:** unfiltered market-cap-weighted index for each CRSP-like universe.
- **Temporary CSI:** `dynamic_csi` score triggers finite exclusions; firms can re-enter after lockout expiry.
- Temporary thresholds: $CV\ FPR \leq 1\%$, $CV\ FPR \leq 3\%$, and Youden J .
- Temporary lockouts: 1, 2, 3, and 5 years; this gives 12 filtered strategies per universe.
- **Permanent CSI:** `permanent_csi` score triggers absorbing removal after first active signal.
- Permanent thresholds: $CV\ FPR \leq 1\%$, $CV\ FPR \leq 3\%$, and Youden J ; this gives 3 filtered strategies per universe.

Portfolio rule

$$E_{i,t}^{temp} = \mathbf{1}\{\hat{p}_{i,t} \geq \tau\} \times \mathbf{1}\{t \leq t_0 + L\},$$

$$E_{i,t}^{perm} = \mathbf{1}\{\exists s \leq t : \hat{p}_{i,s} \geq \tau\},$$

$$w_{i,t}^{screen} = \frac{(1 - E_{i,t})MCAP_{i,t}}{\sum_{j \in U_t} (1 - E_{j,t})MCAP_{j,t}}.$$

- Thresholds are calibrated from model validation/CV predictions, not from index OOS returns.
- Best-rule appendix rows select the highest annualized geometric return within each track, period, and universe.
- Error-cost decomposition reports annualized pp contributions from false positives, false negatives, true positives, and true negatives.

Appendix A15: Temporary CSI Index Results - Test

Performance: benchmark row followed by selected best rule

Universe	Strategy	Geo ret.	Ann. SD	Sharpe	Max DD	ES 2.5%	Δ pp
Total Market	Benchmark	13.75%	12.05%	0.889	-14.62%	-8.44%	-
	Youden / 3yr	14.43%	11.08%	1.011	-12.57%	-7.51%	+0.676
Large Cap	Benchmark	14.18%	11.60%	0.952	-13.82%	-8.02%	-
	Youden / 3yr	14.68%	10.88%	1.047	-12.09%	-7.19%	+0.493
Mid Cap	Benchmark	13.82%	12.85%	0.846	-15.30%	-8.68%	-
	Youden / 3yr	14.27%	11.91%	0.937	-13.70%	-8.05%	+0.449
Small Cap	Benchmark	11.44%	15.55%	0.586	-20.07%	-10.80%	-
	Youden / 2yr	12.13%	14.03%	0.678	-18.27%	-10.24%	+0.692

Error-cost contribution of selected best rule

Universe	Best rule	FP cost	FN cost	TP gain	TN gain
Total Market	Youden / 3yr	-1.528	-0.001	+0.084	+2.097
Large Cap	Youden / 3yr	-1.212	+0.000	+0.003	+1.715
Mid Cap	Youden / 3yr	-1.810	+0.000	+0.015	+2.651
Small Cap	Youden / 2yr	-3.224	-0.008	+0.360	+3.130

Appendix A16: Temporary CSI Index Results – Test-OOS

Performance: benchmark row followed by selected best rule

Universe	Strategy	Geo ret.	Ann. SD	Sharpe	Max DD	ES 2.5%	Δ pp
Total Market	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	–
	FPR3 / 5yr	13.41%	18.50%	0.615	-24.20%	-11.44%	+0.119
Large Cap	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	–
	FPR1 / 1yr	14.26%	18.49%	0.656	-24.79%	-10.83%	-0.020
Mid Cap	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	–
	Youden / 1yr	9.68%	20.09%	0.414	-23.55%	-13.03%	+0.030
Small Cap	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	–
	Youden / 3yr	9.03%	21.98%	0.370	-29.12%	-15.60%	+1.372

Error-cost contribution of selected best rule

Universe	Best rule	FP cost	FN cost	TP gain	TN gain
Total Market	FPR3 / 5yr	-0.970	-0.003	+0.064	+0.782
Large Cap	FPR1 / 1yr	-0.018	+0.000	+0.000	+0.005
Mid Cap	Youden / 1yr	-3.552	+0.000	-0.004	+2.875
Small Cap	Youden / 3yr	-7.648	-0.034	+0.463	+7.455

Appendix A17: Permanent CSI Index Results – Test

Performance: benchmark row followed by selected best rule

Universe	Strategy	Geo ret.	Ann. SD	Sharpe	Max DD	ES 2.5%	Δ pp
Total Market	Benchmark	13.75%	12.05%	0.889	-14.62%	-8.44%	–
	Youden / perm.	14.19%	10.39%	1.050	-10.44%	-7.06%	+0.432
Large Cap	Benchmark	14.18%	11.60%	0.952	-13.82%	-8.02%	–
	FPR3 / perm.	14.40%	11.44%	0.981	-13.64%	-7.91%	+0.220
Mid Cap	Benchmark	13.82%	12.85%	0.846	-15.30%	-8.68%	–
	Youden / perm.	14.63%	10.80%	1.050	-12.15%	-7.45%	+0.805
Small Cap	Benchmark	11.44%	15.55%	0.586	-20.07%	-10.80%	–
	Youden / perm.	13.09%	13.27%	0.774	-17.90%	-9.90%	+1.652

Error-cost contribution of selected best rule

Universe	Best rule	FP cost	FN cost	TP gain	TN gain
Total Market	Youden / perm.	-4.490	-0.003	+0.055	+4.367
Large Cap	FPR3 / perm.	-0.429	+0.000	-0.000	+0.510
Mid Cap	Youden / perm.	-4.871	+0.000	-0.001	+5.308
Small Cap	Youden / perm.	-5.048	-0.020	+0.209	+5.617

Appendix A18: Permanent CSI Index Results – Test-OOS

Performance: benchmark row followed by selected best rule

Universe	Strategy	Geo ret.	Ann. SD	Sharpe	Max DD	ES 2.5%	Δ pp
Total Market	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	–
	FPR3 / perm.	13.76%	18.64%	0.629	-24.21%	-11.38%	+0.472
Large Cap	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	–
	FPR3 / perm.	14.62%	18.30%	0.678	-24.10%	-10.73%	+0.345
Mid Cap	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	–
	FPR1 / perm.	10.08%	20.56%	0.427	-24.38%	-13.85%	+0.426
Small Cap	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	–
	FPR3 / perm.	7.61%	23.13%	0.306	-29.39%	-16.11%	-0.042

Error-cost contribution of selected best rule

Universe	Best rule	FP cost	FN cost	TP gain	TN gain
Total Market	FPR3 / perm.	-1.261	-0.001	+0.022	+1.371
Large Cap	FPR3 / perm.	-0.849	+0.000	+0.000	+0.969
Mid Cap	FPR1 / perm.	-0.167	+0.000	+0.000	+0.536
Small Cap	FPR3 / perm.	-3.109	-0.017	+0.076	+2.938

Bibliography I

- Edward I. Altman. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4):589–609, 1968.
- John Y. Campbell, Jens Hilscher, and Jan Szilagyi. In search of distress risk. *The Journal of Finance*, 63(6):2899–2939, 2008.
- Michael Cembalest. The agony & the ecstasy: The risks and rewards of a concentrated stock position. Eye on the market special edition, J.P. Morgan Asset Management, 2014. September 2, 2014.
- Stephen Penman and Francesco Reggiani. Fundamentals of value versus growth investing and an explanation for the value trap. *Financial Analysts Journal*, 74(4):103–119, 2018. doi: 10.2469/faj.v74.n4.6.
- Tyler Shumway. The delisting bias in CRSP data. *The Journal of Finance*, 52(1):327–340, 1997.
- Tyler Shumway. Forecasting bankruptcy more accurately: A simple hazard model. *The Journal of Business*, 74(1):101–124, 2001.
- Zaki Tewari, Michal Galas, and Philip Treleaven. Predicting catastrophic stock implosions: A machine learning approach. Technical report, University College London, 2024. Working Paper.